

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T48 · Right-Angled Triangles: Pythagoras and Trigonometry

With Answers

Questions with full worked solutions

■ 45 pages

FAMILY MAP

Right Angled Triangles Pythagoras Trigonometry

F01	Resolve multi-step two-dimensional right-triangle models	9 Q 35 marks
F02	Use right-triangle trigonometry for lengths	6 Q 18 marks
F03	Use Pythagoras in two dimensions	2 Q 8 marks
F04	Use angles of elevation and depression	4 Q 14 marks

Working Unit 17

4MA1-2H Q10

ORIGINAL QUESTION**5 marks**

- 10 ABD and ABC are right-angled triangles.

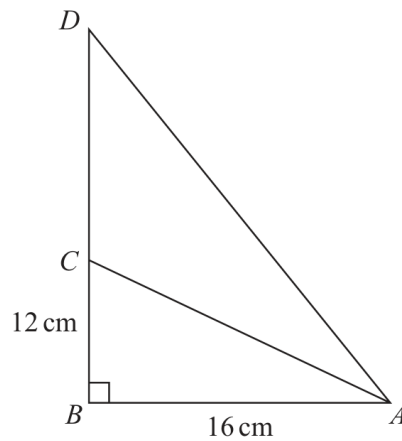


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

Worked Solution 17

Connected right-angled triangles

UNIT 1
5 marks**Small triangle**

$$AC = \sqrt{16^2 + 12^2} = 20 \text{ cm.}$$

Large hypotenuse and height

$$AD = 1.5(20) = 30, \quad BD = \sqrt{30^2 - 16^2} = \sqrt{644}.$$

Required segment

$$CD = BD - BC = \sqrt{644} - 12 = 13.3772 \dots$$

Final answer

$$CD = 13.4 \text{ cm (3 s.f.)}$$

Question 07

4MA1-1H Q11

ORIGINAL QUESTION

4 marks

- 11 The diagram shows a roof support.

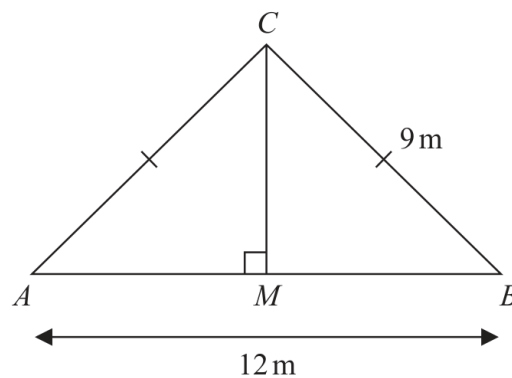


Diagram **NOT**
accurately drawn

The roof support is made from four lengths of wood, AB , AC , BC and MC

$$AC = BC = 9 \text{ m} \quad AB = 12 \text{ m}$$

$$\text{angle } AMC = 90^\circ$$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.

Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.

Show your working clearly.

..... euros

Worked Solution 07

Pythagoras and purchase length

MODULAR UNIT 1

4 marks

Central support

$$AM = 6, \quad CM = \sqrt{9^2 - 6^2} = \sqrt{45} = 6.708\dots$$

Whole metres

$$12 + 9 + 9 + 7 = 37 \text{ m}; \quad 37(21.50) = 795.50.$$

Final answer

EUR 795.50

Question 19

4MA1-2H Q5

ORIGINAL QUESTION

3 marks

- 5 The diagram shows triangle ABC

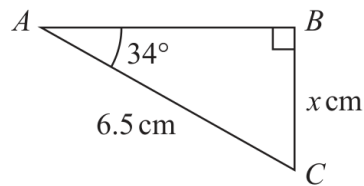


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Worked Solution 19

Right-triangle trigonometry

MODULAR UNIT 1

3 marks

Sine

$$\sin 34^\circ = x/6.5 \Rightarrow x = 6.5 \sin 34^\circ.$$

Final answer

3.6 cm

Question 15

4MA1-2H Q2 whole

ORIGINAL QUESTION UNIT 1

4 marks

- 2 The diagram shows triangle ABC and triangle ECD

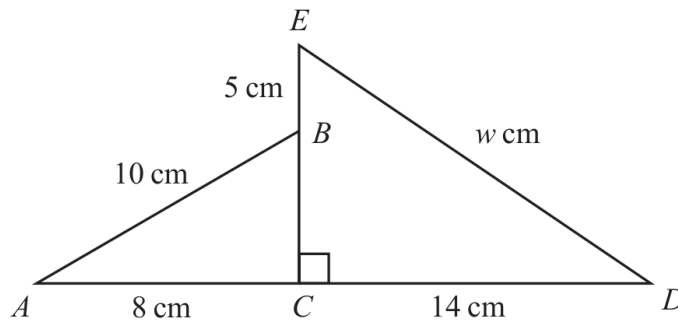


Diagram **NOT**
accurately drawn

ACD and EBC are straight lines.

$$AB = 10 \text{ cm} \quad AC = 8 \text{ cm} \quad EB = 5 \text{ cm} \quad CD = 14 \text{ cm} \quad ED = w \text{ cm}$$

Work out the value of w

Give your answer correct to one decimal place.

$w = \dots\dots\dots$

Worked solution 15

Two right triangles joined at C

MODULAR UNIT 1

4 marks

whole – Two right triangles joined at C

Find BC

$$BC = \sqrt{10^2 - 8^2} = \sqrt{36} = 6.$$

Find EC

$$EC = EB + BC = 5 + 6 = 11.$$

Use the second right triangle

$$w^2 = ED^2 = EC^2 + CD^2 = 11^2 + 14^2 = 317.$$

Round the length

$$w = \sqrt{317} = 17.804\dots \approx 17.8 \text{ cm.}$$

Final answer

w = 17.8 cm

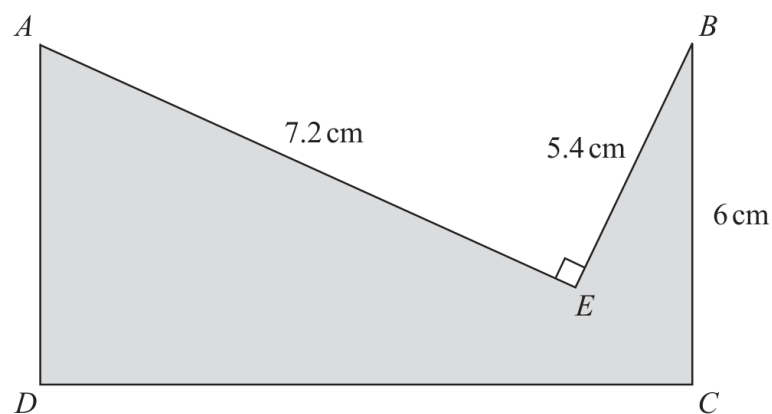
Question 01

4MA1-1H Q5

ORIGINAL QUESTION

4 marks

- 5 The diagram shows a shaded shape $AEBCD$ made by removing triangle AEB from rectangle $ABCD$



$$AE = 7.2 \text{ cm} \quad BE = 5.4 \text{ cm} \quad BC = 6 \text{ cm} \quad \text{angle } AEB = 90^\circ$$

Work out the perimeter of the shaded shape.

..... cm

Worked Solution 01

Perimeter with Pythagoras

MODULAR UNIT 1

4 marks

Find the missing rectangle side

$$AB = \sqrt{7.2^2 + 5.4^2} = 9.$$

Add the shaded boundary

$$7.2 + 5.4 + 6 + 9 + 6 = 33.6.$$

Final answer

33.6 cm

Question 17

4MA1-2H Q12

ORIGINAL QUESTION

4 marks

12 The diagram shows right-angled triangle ABD

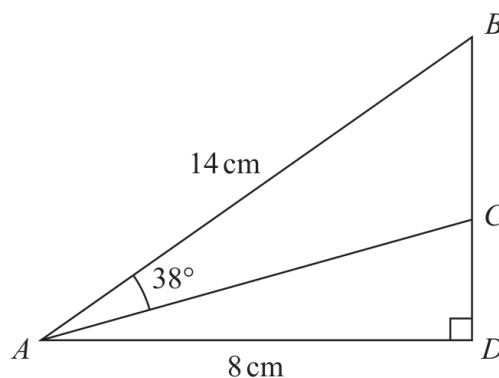


Diagram **NOT**
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

C is the point on BD such that angle $BAC = 38^\circ$

Work out the length of CD

Give your answer correct to 3 significant figures.

..... cm

Worked Solution 17

Two right triangles and a split hypotenuse

MODULAR UNIT 1

4 marks

Find the angle in ABD

$$\cos(\angle BAD) = \frac{AD}{AB} = \frac{8}{14} \Rightarrow \angle BAD \approx 55.15^\circ.$$

Find the angle in ACD

$$\angle CAD = 55.15 - 38 \approx 17.15^\circ.$$

Find CD

$$\tan(17.15^\circ) = CD/8 \Rightarrow CD \approx 2.47 \text{ cm}.$$

Final answer

2.47 cm

Original question 09

4MA1-1H Q12 M01

MODULAR UNIT 1

4 marks

- 12 ABC is a right-angled triangle.
 D is a point on BC

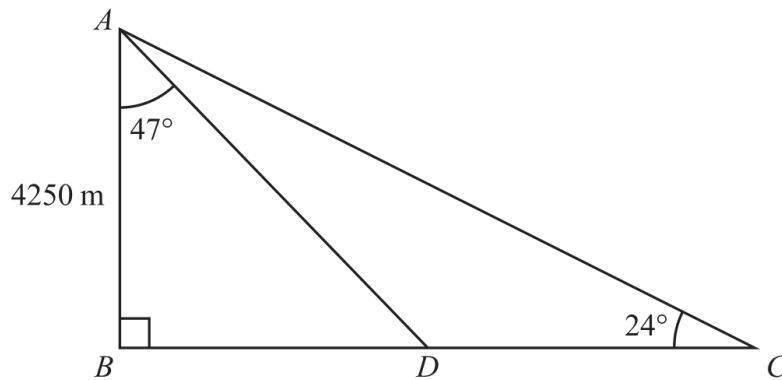


Diagram **NOT**
accurately drawn

$$AB = 4250 \text{ m} \quad \text{angle } BAD = 47^\circ \quad \text{angle } BCA = 24^\circ$$

Work out the length of DC

Give your answer correct to the nearest integer.

..... m

Worked solution 09

Find DC using trigonometry

MODULAR UNIT 1

4 marks

M01 – Find DC using trigonometry**Find BD**

$$BD = 4250 \tan 47^\circ = 4557.567 \dots$$

Find BC

$$BC = 4250 / \tan 24^\circ = 9545.656 \dots$$

Subtract

$$DC = BC - BD = 4988.089 \dots \text{ m.}$$

Round

$$DC = 4988 \text{ m.}$$

Final answer

4988 m

Original question 20

4MA1-2H Q3 M01

MODULAR UNIT 1

3 marks

- 3 The diagram shows triangle PQR

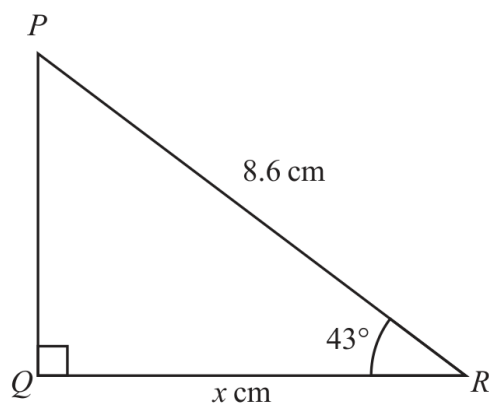


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Worked solution 20

Use cosine in a triangle

MODULAR UNIT 1

3 marks

M01 – Use cosine in a triangle**Choose the ratio**

$$\cos 43^\circ = x/8.6.$$

Rearrange

$$x = 8.6 \cos 43^\circ = 6.300 \dots$$

Round

$$x = 6.3 \text{ cm.}$$

Final answer

6.3 cm

Original question 19

4MA1-2H Q6 Q6

ORIGINAL QUESTION**3 marks**

- 6 The diagram shows triangle PQR .

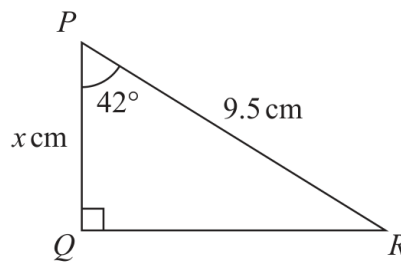


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Worked solution 19

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q6 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Use right-triangle trigonometry for lengths • Variant: Find adjacent side with cosine or tangent

MethodPR is the hypotenuse and PQ = x is adjacent to 42 degrees, so $\cos 42 \text{ degrees} = x/9.5$.*Why: Use cosine for adjacent over hypotenuse.***Method** $x = 9.5 \cos 42 \text{ degrees} = 7.059... .$ *Why: Rearrange and evaluate in degree mode.***Method**Therefore $x = 7.1$ to one decimal place.*Why: Round once at the requested accuracy.***Final answer****x:** $x = 7.1$

Original question 04

4MA1-1H Q3

ORIGINAL QUESTION**3 marks**

- 3 The diagram shows triangle ABC

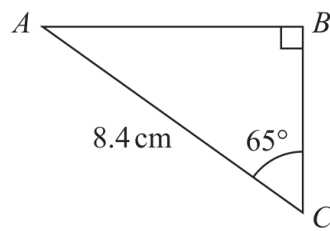


Diagram **NOT**
accurately drawn

Work out the length of the side AB
Give your answer correct to 3 significant figures.

..... cm

Worked solution 04

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q3 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Use right-triangle trigonometry for lengths • Variant: Find opposite side with sine or tangent

Method

$$AB/8.4 = \sin 65^\circ / \sin 90^\circ.$$

*Why: Use the sine rule in the right triangle.***Method**

$$AB = 8.4 \sin 65^\circ.$$

*Why: $\sin 90^\circ = 1$, so rearrange for AB.***Method**

$$AB = 7.61 \text{ cm (3 s.f.)}.$$

*Why: Round the calculator value 7.612... to 3 significant figures.***Final answer****AB: 7.61 cm**

Original question 24

4MA1-2H Q14

ORIGINAL QUESTION**3 marks**

- 14 A zip wire is shown as the dashed line AC in the diagram.

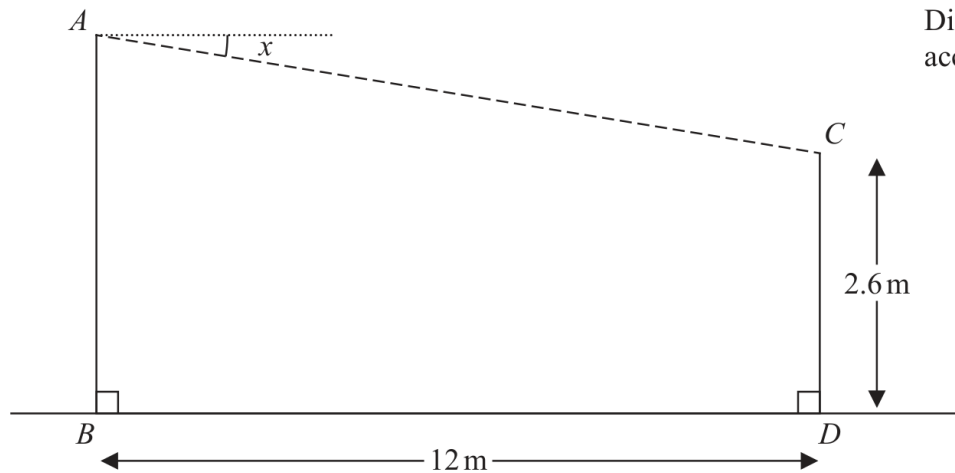


Diagram **NOT**
accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$$CD = 2.6 \text{ m} \quad BD = 12 \text{ m}$$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

Worked solution 24

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q14 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Resolve multi-step two-dimensional right-triangle models • Variant: Use a right-triangle result in a practical threshold or cost

Method

The vertical height difference is $y = 12 \tan 5^\circ = 1.04986... \text{ m}$.

Why: Use the right triangle between the posts.

Method

$AB = 2.6 + y = 3.64986... \text{ m}$.

Why: Add the height of CD.

Method

The least possible height is 3.65 m (3 s.f.).

Why: Round the result.

Final answer

AB: 3.65 m

9 Here is a right-angled triangle.

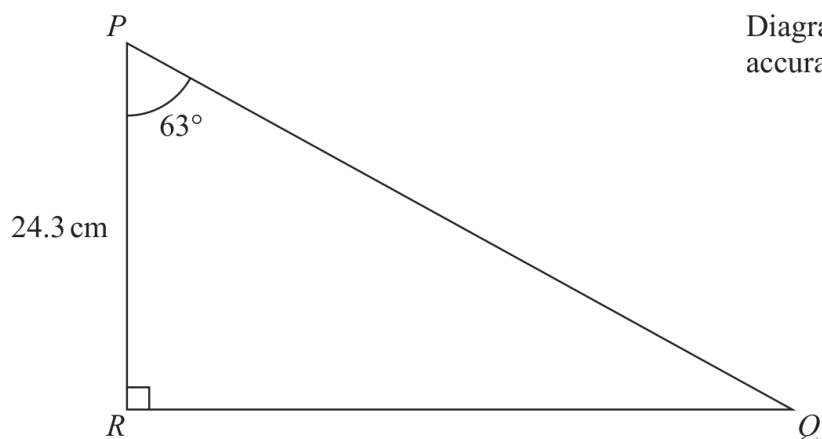


Diagram **NOT**
accurately drawn

Calculate the length of PQ .
Give your answer correct to 3 significant figures.

..... cm

Worked solution

January 2020 | Question 9 | 3 marks

Family: Use right-triangle trigonometry for lengths • Variant: Find opposite side with sine or tangent

Method / working

1. $\cos 63^\circ = 24.3/PQ$.

2. $PQ = 24.3/\cos 63^\circ = 53.525\dots$ cm.

3. To 3 s.f., $PQ = 53.5$ cm.

Final answer: 53.5 cm

10 Here is isosceles triangle ABC .

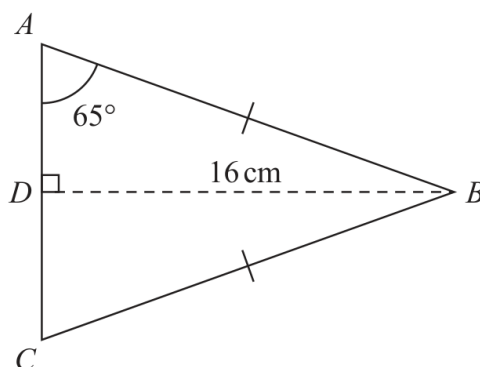


Diagram **NOT**
accurately drawn

D is the midpoint of AC and $DB = 16$ cm.

Angle $DAB = 65^\circ$

Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

Worked solution

January 2021 | Question 10 | 4 marks

Family: Resolve multi-step two-dimensional right-triangle models • Variant: Combine right-triangle trig with area or perimeter

Method / working

1. $AB = 16 / \sin 65^\circ = 17.654 \dots$ cm.
2. $AD = 16 \tan 25^\circ = 7.460 \dots$ cm.
3. Because D and midpoint conditions give perimeter $2AB + 2AD$.
4. Perimeter = 50.2 cm to 1 d.p.

Final answer: 50.2 cm

7 Here is a right-angled triangle.

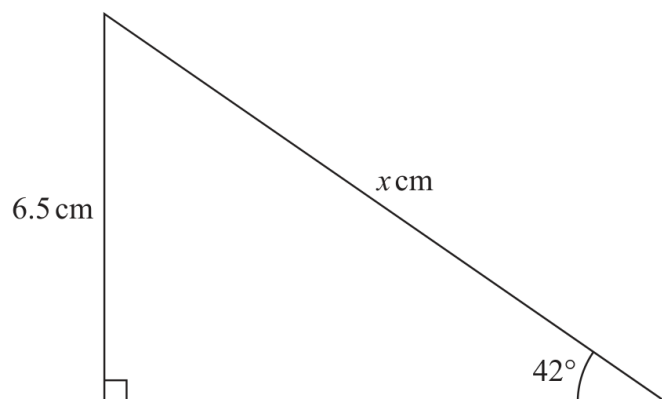


Diagram **NOT**
accurately drawn

Work out the value of x .
Give your answer correct to one decimal place.

$x = \dots\dots\dots$

Worked solution

June 2021 | Question 7 | 3 marks

Family: Use right-triangle trigonometry for lengths • Variant: Find hypotenuse with sine or cosine

Method / working

1. The 6.5 cm side is opposite 42° and x is the hypotenuse.
2. Use $\sin 42^\circ = 6.5/x$, so $x = 6.5/\sin 42^\circ$.
3. Round to 1 decimal place: $x = 9.7$ cm.

Final answer: 9.7 cm

2 Here is a right-angled triangle.

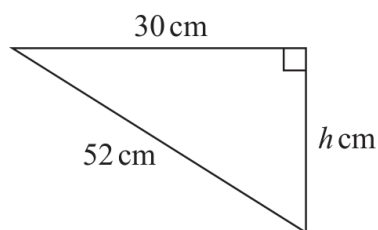


Diagram **NOT**
accurately drawn

Calculate the value of h .
Give your answer correct to 3 significant figures.

$h =$

Worked solution

November 2021 | Question 2 | 3 marks

Family: Use Pythagoras in two dimensions | Variant: Find a shorter side by Pythagoras

METHOD / WORKING

1. $h^2 + 30^2 = 52^2$

2. $h^2 = 52^2 - 30^2 = 1804$

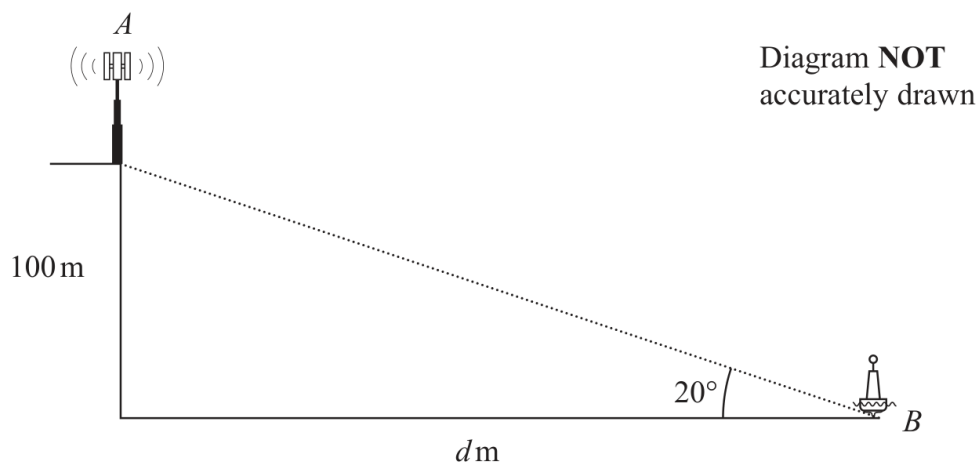
3. $h = \sqrt{1804} = 42.473\dots$

4. Answer: 42.5cm to 3 significant figures.

FINAL ANSWER

42.5 cm to 3 significant figures.

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

Worked solution

November 2021 | Question 12 | 3 marks

Family: Use angles of elevation and depression | Variant: Find horizontal distance from an elevation angle

METHOD / WORKING

1. (a)

$$2. \tan 20^\circ = \frac{100}{d}$$

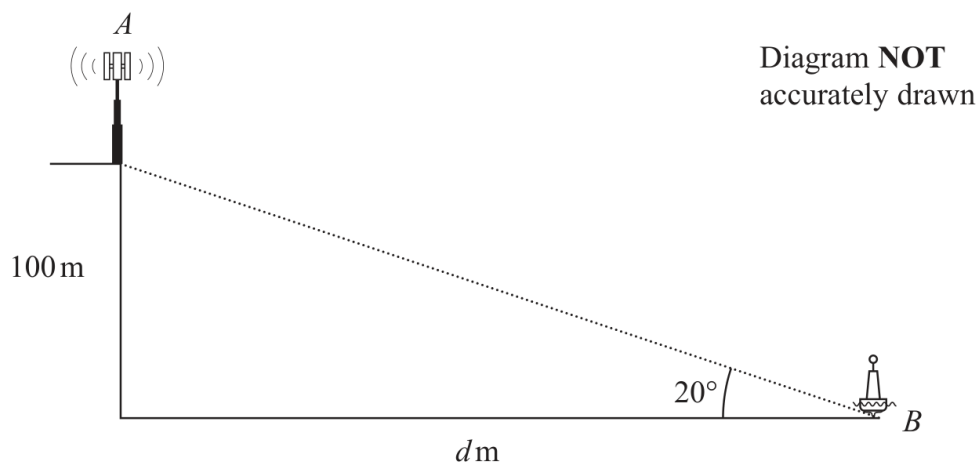
$$3. d = \frac{100}{\tan 20^\circ} = 274.747\dots$$

4. So

5. $d=275$ m**FINAL ANSWER**

(a) 275 m

- 12 The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.



The height of the cliff is 100 metres.

The buoy is at the point B that is d metres from the base of the cliff.

The angle of elevation from B to the top of the cliff is 20°

- (a) Calculate the value of d .

Give your answer correct to 3 significant figures.

$d = \dots\dots\dots$
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.

The angle of elevation from B to A is 25°

- (b) Calculate the height of the radio mast.

Give your answer correct to 3 significant figures.

$\dots\dots\dots$ m
(3)

Worked solution

November 2021 | Question 12 | 3 marks

Family: Resolve multi-step two-dimensional right-triangle models | Variant: Use two connected right triangles

METHOD / WORKING

1. (b) The angle of elevation to the top of the radio mast is 25° .

$$2. \tan 25^\circ = \frac{100 + h}{274.747...}$$

$$3. 100 + h = 274.747... \tan 25^\circ = 128.116...$$

$$4. h = 28.116...$$

5. Answer: (a) $275m$, (b) $28.1m$.

FINAL ANSWER

(b) $28.1m$.

- 7 The diagram shows a right-angled triangle.

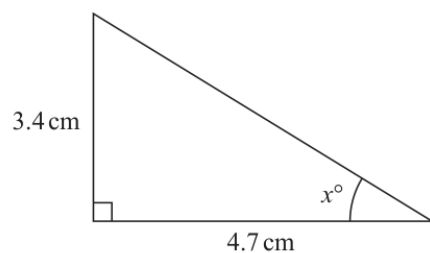


Diagram **NOT**
accurately drawn

Calculate the value of x .
Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

Answer reference | May-Nov 2020 | Question 7

Family: Use angles of elevation and depression • Variant: Find a height from an elevation or depression angle

7	$\tan x = \frac{3.4}{4.7}$ oe eg $\cos x = \frac{4.7}{\sqrt{3.4^2 + 4.7^2}}$ oe			M1	or $\sin x = \frac{3.4 \sin 90}{\sqrt{3.4^2 + 4.7^2}}$ oe
	$(x =) \tan^{-1}\left(\frac{3.4}{4.7}\right)$ oe eg $(x =) \cos^{-1}\left(\frac{4.7}{\sqrt{3.4^2 + 4.7^2}}\right)$			M1	or $(x =) \sin^{-1}\left(\frac{3.4 \sin 90}{\sqrt{3.4^2 + 4.7^2}}\right)$ oe
		35.9	3	A1	accept 35.7 - 36.1
				Total 3 marks	

Worked solution

May-Nov 2020 | Question 7 | 3 marks

Family: Use angles of elevation and depression • Variant: Find a height from an elevation or depression angle

Official final mark-scheme pages are embedded immediately after this question.

- 10 The diagram shows an isosceles triangle, with base length 24 cm.

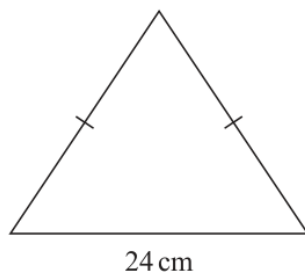


Diagram **NOT**
accurately drawn

The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm²

(Total for Question 10 is 5 marks)

Answer reference | January 2023 | Question 10

Family: Use Pythagoras in two dimensions • Variant: Use Pythagoras inside a composite 2D shape

Question	Working	Answer	Mark	Notes
10	$(54 - 24) \div 2 (=15)$ [may be marked on diagram]		5	M1
	$"15" - (24 \div 2) (= 81)$			M1 ft their "15" (if > 12)
	[height =] $\sqrt{"15" - (24 \div 2)^2} (= 9)$			M1 ft their "15" (if > 12)
	$(24 \times "9") \div 2$ oe			M1 figures must be from correct working
	Correct answer scores full marks (unless from obvious incorrect working)	108		A1 allow 107.9 – 108.1
	ALTERNATIVES BELOW			Total 5 marks
10	$(54 - 24) \div 2 (=15)$ [may be marked on diagram]		5	M1
	or $x = \cos^{-1}\left(\frac{"12"}{"15"}\right) (= 36.86\dots)$ or $y = \sin^{-1}\left(\frac{24 \div 2}{"15"}\right) (= 53.13\dots)$ or $A = \cos^{-1}\left(\frac{15^2 + 15^2 - 24^2}{2 \times 15 \times 15}\right) (= 106.2\dots)$ or $B = \cos^{-1}\left(\frac{15^2 + 24^2 - 15^2}{2 \times 15 \times 24}\right) (= 36.8\dots)$			M1 ft their "15" (if > 12) [using Hero's formula $S = 0.5 \times 54 (= 27)$ and] $27 \times (27 - 24) \times (27 - "15") \times (27 - "15")$
	or "12"tan"36.86..." (= 9) (allow 8.9... for these) or "12" ÷ tan"53.13..." (= 9) or "15" × sin "36.86..." (= 9) or "15" × cos "53.13..." (= 9)			M1 ft their "15" (if > 12) M2 for $0.5 \times 24 \times "15" \times \sin "36.86\dots"$ or $0.5 \times "15" \times "15" \times \sin(2 \times "53.13\dots")$ or $0.5 \times "15" \times "15" \times \sin("106.2\dots")$ or $\sqrt{"27"("27" - 24)("27" - "15")("27" - "15")}$
	$(24 \times "9") \div 2$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	108		A1 allow 107.9 – 108.1
				Total 5 marks

Worked solution

January 2023 | Question 10 | 5 marks

Family: Use Pythagoras in two dimensions • Variant: Use Pythagoras inside a composite 2D shape

Official final mark-scheme pages are embedded immediately after this question.

11 ABC is a right-angled triangle.

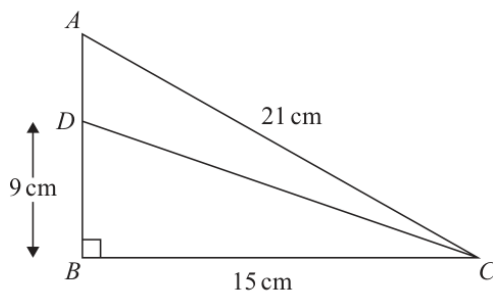


Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$

Work out the size of angle ACD

Give your answer correct to one decimal place.

(Total for Question 11 is 4 marks)

Worked solution

November 2025 | Question 11 | 4 marks

Family: Use angles of elevation and depression | Variant: Find a height from an elevation or depression angle

METHOD / WORKING

Worked solution
November 2025 Question 11 4 marks
Family: Use angles of elevation and depression Variant: Find a height from an elevation or depression angle
Solution
First find AB:
$AB^2 = 21^2 - 15^2 = 216$
$AB = 14.696\dots$
$AD = AB - DB = 14.696\dots - 9 = 5.696\dots$
Angle ACB:
$\cos ACB = (15)/(21)$
$ACB = 44.415\dots^\circ$
Angle DCB:
$\tan DCB = (9)/(15)$
$DCB = 30.963\dots^\circ$
$ACD = 44.415\dots - 30.963\dots = 13.451\dots^\circ$
Answer: 13.5° .
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

11 The diagram shows a right-angled triangle ABC

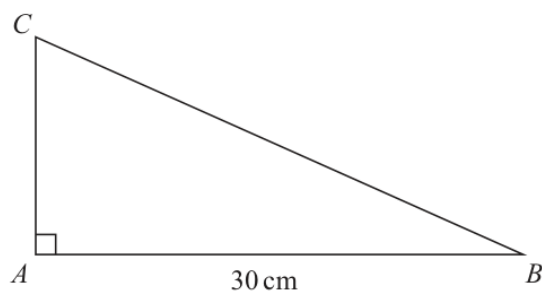


Diagram **NOT**
accurately drawn

Area of triangle $ABC = 240 \text{ cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

Worked solution

November 2025 | Question 11 | 4 marks

Family: Use angles of elevation and depression | Variant: Find a height from an elevation or depression angle

METHOD / WORKING

Worked solution
November 2025 Question 11 4 marks
Family: Use angles of elevation and depression Variant: Find a height from an elevation or depression angle
Solution
Use the area to find AC:
$\frac{1}{2} \times 30 \times AC = 240$
$15AC = 240$
$AC = 16$
Then
$BC^2 = 30^2 + 16^2 = 1156$
$BC = 34$
Perimeter:
$30 + 16 + 34 = 80$
Answer: 80 cm.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER